Winglet Influence on the Aerodynamic Performance of a Finite Wing

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This experiment measured the lift coefficient C_L, drag coefficient C_D, of the Eppler 212 airfoil with multiple winglet setups in a low-speed wind tunnel. Tests were completed at two Reynolds number $Re \approx 150,000$ and $Re \approx 300,000$, from an angle of attack of $-15° \leq a \leq 20°$. With the tunnel turned off, gravitational effects were found by measuring the axial force F_A and normal force F_N at all angles of attack and then removed from the aerodynamic force data. Using measured pressures, and forces when at respective Reynolds numbers, the Lift and Drag coefficients were found for multiple types of winglet families that varied in location, length, and angle, along with a baseline no winglet setup. Overall, winglets increased the lift and reduced induced drag while the impact of winglets varied due to design and was more impactful at the higher Reynolds number. When compared to the baseline airfoil with no winglet, the winglet cases showed greater performance gains with the addition of small additions to the wing.

I. Nomenclature

α	=	Angle of attack (deg)
ρ	=	Air density in test section (kg/m^3)
μ	=	Dynamic viscosity of air (kg/m*s)
μ_{ref}	=	Reference viscosity for Sutherland's law (kg/m*s)
T	=	Air temperature (K)
T_{ref}	=	Reference Air temperature for Sutherland's law (K)
p	=	Static Pressure (Pa)
Δp	=	Dynamic pressure from pitot tube (Pa)
R	=	specific gas constant for air (287 J/kg*K)
V_∞	=	Freestream velocity (m/s)
Re	=	Reynolds number
c	=	Chord length (m)
b	=	span of the wing (m^2)
S	=	Planform area of the wing (m^2)
$F_{A,meas}$	=	Measured axial force from the force balance (N)
$F_{N,meas}$	=	Measured normal force from the force balance (N)
$F_{A,grav}$	=	Axial gravitational offset force (N)
$F_{N,grav}$	=	Normal gravitational offset force (N)
$F_{A,corr}$	=	Gravity corrected axial force (N)
$F_{N,corr}$	=	Gravity corrected Normal force (N)
L	=	Lift force (N)
D	=	Drag force (N)
q	=	dynamic pressure (Pa)

C_L	=	lift coefficient
D_L	=	Drag coefficient

II. **Introduction**

Understanding how lift and drag will act on a finite wing is important in aerodynamic design. When wind moves around the end of a finite wing, strong tip vortices form and create downwash behind the wing. This downwash begins to slightly tilt the lift vector and causes induced drag. In front of the trailing edge of the wing the air will be pulled upward forming upwash that will change the angle of attack. These effects create induced drag which will reduce the aerodynamic efficiency of the wing, especially at low speeds with greater vortex strength [1].

The Eppler 212 finite wing was used in this experiment to see how lift and drag changed with angle of attack as well how the aerodynamic coefficients were affected by different winglet design. Winglets are small vertical or angled surfaces attached to the wingtip that help reduce the strength of the tip vortices. By limiting spanwise flow and changing how the pressure field wraps around the tip, winglets are a great tool to reduce induced drag and can be used to increase lift. Their effectiveness and change are due to where they are placed, how long they are, and the angle they are attached at [1].

To measure the Aerodynamic forces produced by the wing and its winglets, a force balance was used inside the wind tunnel to the model by converting small deflections to electric signals. The balance is also calibrated so the voltages will be converted to physical forces axial F_A and Normal F_N [2].

In this experiment several winglet families were compared to a baseline wing with no winglets attached. Each family varied in location, length, or attachment angle, show a direct assessment of how winglet changes affect the wing's aerodynamic behavior and efficiency. By measuring the lift and drag forces across a range of angles of attack and applying corrections for gravity. The Experiment shows how a small modification at the wingtip of a finite airfoil can influence downwash reduce induced drag and improve the overall efficiency.

III. **Procedure**

The experiment began by mounting the Eppler 212 finite wing to the test section of the low-speed wind tunnel. Next set up a pitot tube in the test section for freestream measurements, a pressure transducer to convert the difference in pressures to voltages, a force balance that measured the axial F_A and Normal F_N forces acting on the Airfoil.

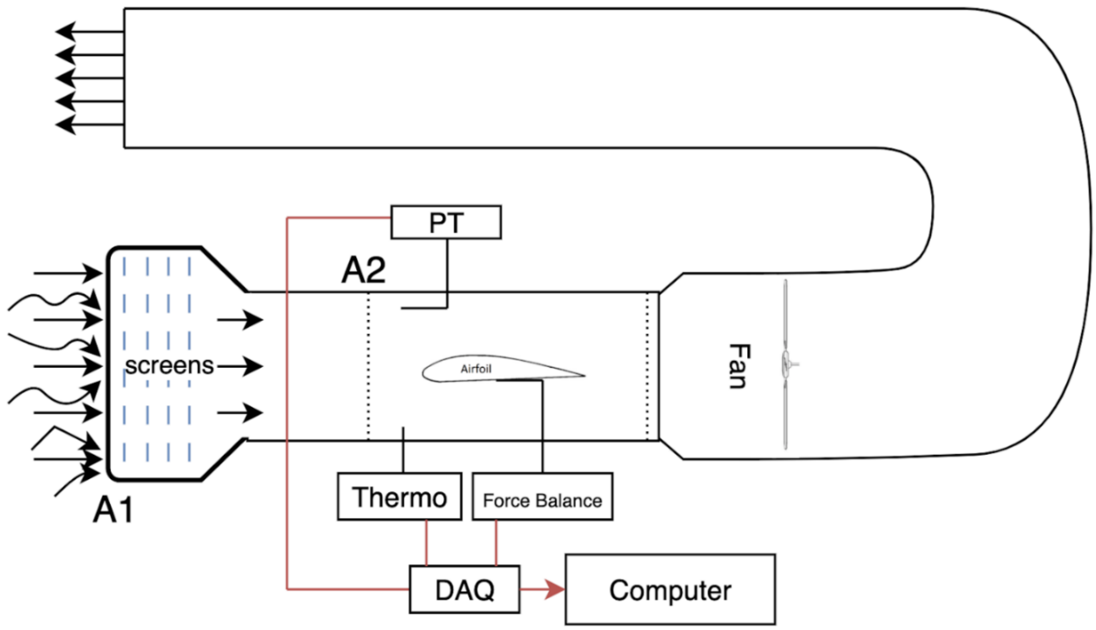


Fig. 1 Experiment setup

Before turning on the wind tunnel the gravitational forces acting on the Eppler 212 airfoil needed to be measured at all angles of attack from $-15°$ to $20°$ and account for the setups offset angle when recording data. The record values represent the weight projected on the force balance and were later subtracted from the aerodynamic forces when the wind tunnel was running as seen in Eqs. (1)-(2). This gives us the ability to measure the true aerodynamic coefficients without the effects of gravity.

$$F_{A,\mathrm{corr}} = F_{A,\mathrm{meas}} - F_{A,\mathrm{grav}} \tag{1}$$

$$F_{N,\mathrm{corr}} = F_{N,\mathrm{meas}} - F_{N,\mathrm{grav}} \tag{2}$$

After zeroing the model turns on the wind tunnel and set to a Reynolds number of aprroximetly $Re \approx 150{,}000$ and $Re = 300{,}000$. The freestream velocity V_∞ was determioned using the pitot tube measurements and Bernoulli's equation Eq. (3).

$$V_\infty = \sqrt{\frac{2\Delta p}{\rho}} \tag{3}$$

Where Δp is the meausured dynamic pressure from the pitot tube. The density ρ of the air in the test section was computed using the ideal gas relation Eq. (4).

$$\rho = \frac{p}{RT} \tag{4}$$

The viscosity μ was found using Sutherland's law Eq. (5) with the constants $\mu_{ref} = 1.716 \times 10^{-5},\ S_\mu = 110.4\ K,$ and $T_{ref} = 273.15$.

$$\mu = \mu_{ref} \left(\frac{T}{T_{ref}}\right)^{\frac{3}{2}} \frac{T_{ref} + S_\mu}{T + S_\mu} \tag{5}$$

Once all calibration data was colleted the airfoil was swept from $-15°$ to $20°$. At each angle the force balance was recorded the aerodynamic forces with the gravtitional offsets already calibrated. The lift and drag were obtained using Eqs. (6)-(7).

$$L = F_{N,\text{corr}} \cos\alpha - F_{A,\text{corr}} \sin\alpha \tag{6}$$
$$D = F_{A,\text{corr}} \cos\alpha + F_{N,\text{corr}} \sin\alpha \tag{7}$$

Next nodimensionalized, then lift and drag coefficents were calculated using Eq. (7)-(8).

$$C_L = \frac{L}{\frac{1}{2}\rho V_\infty^2 S} \tag{7}$$
$$C_D = \frac{D}{\frac{1}{2}\rho V_\infty^2 S} \tag{8}$$

Each winglet configuration was tested using the same process so the lift curves and drag polars could be compared directly.

Data was processed in MATLAB, for each winglet configuration and Reynolds number, the code first reads the appropriate test file and extracts the pitch angle, normal force, and axial force colums. Then the gravity correction data was fit with a second order polynomial and was subtracted from the aerodynamic data which produced the corrected axial and normal forces used in the lift and drag equations. The next program asigned the ambient tempetrue and pressure for each eonfiguration and computes the air density from ideal gas law, which then calculaates the dynamic viscosity with suterlands law. With the known Reynolds number the fresstream velocity was found and used to find the dynamic pressure. Next Eqs. (6)-(7) were used to find L and D which were later nondimenzionled to get the lift and drag coefficients. uncertainties where found by propagating the measurment errors as seen in Eqs. (9)-(10)

$$\delta C_L = \sqrt{\left(\frac{\partial C_L}{\partial F_N}\,\delta F_N\right)^2 + \left(\frac{\partial C_L}{\partial F_A}\,\delta F_A\right)^2 + \left(\frac{\partial C_L}{\partial q}\,\delta q\right)^2 + \left(\frac{\partial C_L}{\partial \alpha}\,\delta \alpha\right)^2} \tag{9}$$
$$\delta C_D = \sqrt{(\frac{\partial C_D}{\partial F_N}\,\delta F_N)^2 + (\frac{\partial C_D}{\partial F_A}\,\delta F_A)^2 + (\frac{\partial C_D}{\partial q}\,\delta q)^2 + (\frac{\partial C_D}{\partial \alpha}\,\delta \alpha)^2} \tag{10}$$

Lastly the script orders the results in three tables for density and dynamic pressure, C_L vs α data, and drag polar data then creates 12 plots with all the table data.

IV. **Results**

When comparing the lift curves to the winglet location family patterns emerged. The front and mid mounted winglets produced higher lift than the back and baseline winglet at most angle of attacks. This is due to the location where the vortex begins to form at the tip of the wing. The back winglet only slightly deviated from the baseline setup, but all curves followed a linear trend throughout all angles, with the biggest difference appearing around the stall points. These differences also appear in the drag polar plots. Both the mid and front positions shift the polar to the left of the baseline curve, indicating reduced induced drag for the given lift. Meanwhile the back position shows minimal drag reduction and lift improvement. As lift gets higher, greater separation begins to form between the front and middle when compared to the back and baseline. Overall, the location family shows that winglet placement can strongly influence vortex control with the front and middle position being the most impactful for aerodynamic benefits.

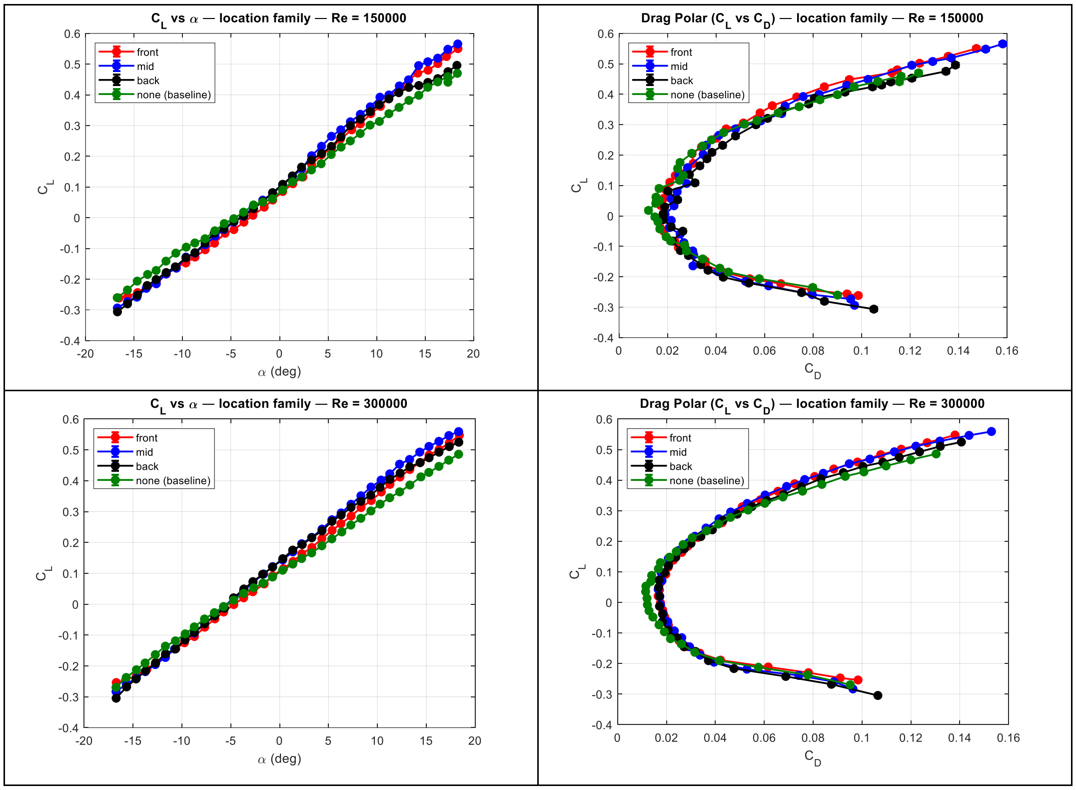


Fig. 2 Winglet Location Family (front, middle, back, none) C_L vs α and C_L vs C_D at Re=150,000 and Re=300,000.

Analyzing the Winglet length family a noticeable change occurs as the winglet grows longer. Increasing the length of the winglet resulted in a clear and consistent increase in lift on the plots. This aligns with the expectations that the longer winglets maintained higher lift values even while approaching the stall while the small and medium winglets provided modest improvements. The patterns align very closely with the drag polars with the long winglet producing the greatest lift and the small winglet being similar to the baseline setup. The long winglet clearly gives a strong reduction to tip vortex strength. This family makes it clear that winglet length has a strong impact on both lift and induced drag.

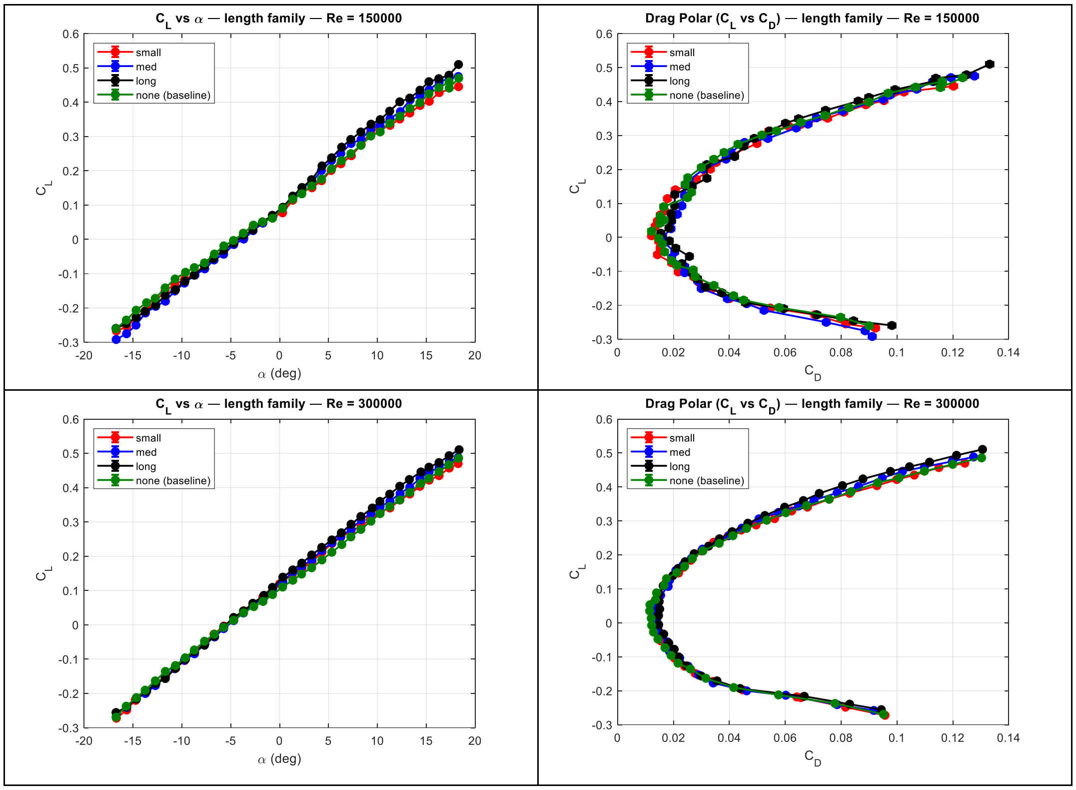
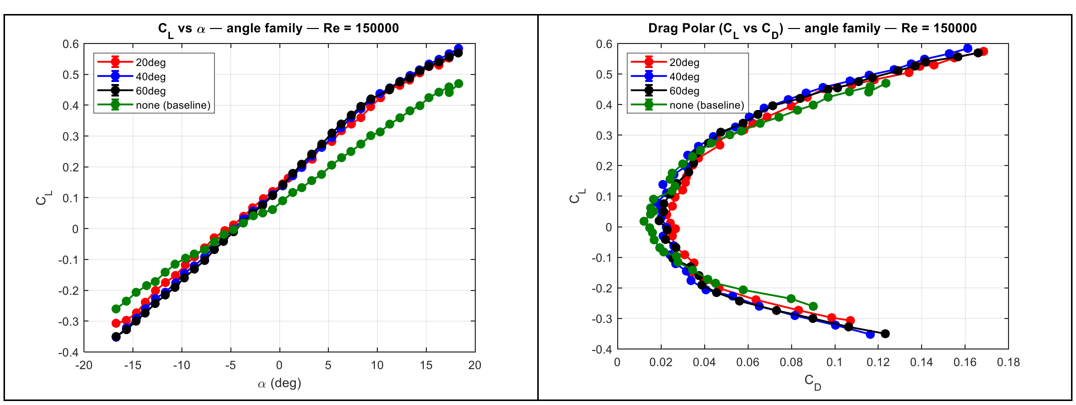


Fig. 3 Winglet Length Family (small, medium, long, none) C_L vs α and C_L vs C_D at Re=150,000 and Re=300,000.

The winglet angle family highlights how the angle influences the aerodynamic effects. The lift curve shows the greatest deviation from the baseline with all angles providing much higher lift. In all plots the twenty-degree and fourty-degree angle winglets produced the highest lift while the sixty-degree angle gave less at higher angles of attack. This can be due to the low angles of winglets aligning with the direction of flow allowing for reduced roll up. In the drag polars the twenty-degree configuration gives the greatest drag while the fourty-degree winglet exhibits the least. This shows that high winglets at higher angles are ineffective since they achieve the lowest lift and a high drag. In summary, the angle family of winglets is important for lift generation at the price of a greater drag.



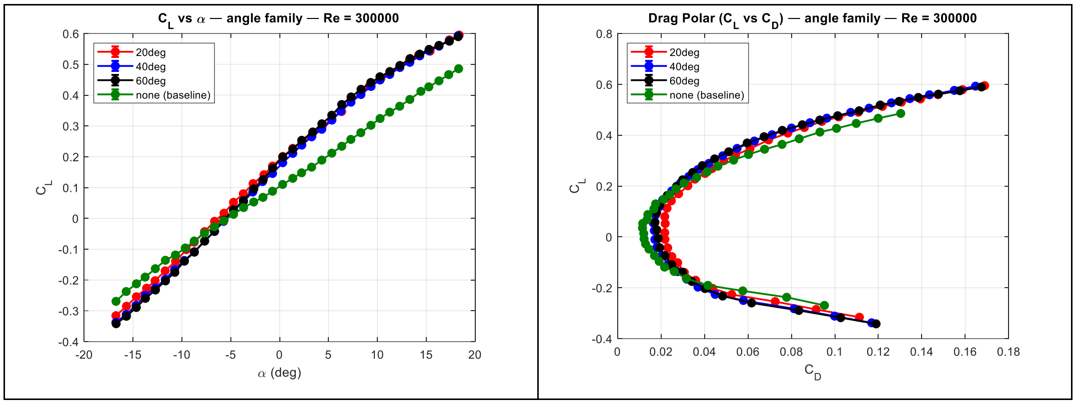


Fig. 4 Winglet Angle Family (20°, 40°, 60°, none) C_L vs α and C_L vs C_D at Re=150,000 and Re=300,000.

V. Conclusion

This experiment helped measured how different winglet designs can influence the aerodynamic coefficients C_L and C_D of the finite Eppler 212 wing by measuring lift and drag at two different Reynolds numbers 150,000 and 300,000 and a range of angles of attack. The behavior aligned with what was expected of a finite wing in theory, with the winglets increasing the lift when compared to the baseline test results, especially at high angles of attack. This is because winglets help weaken the wingtip vortices and downwash, allowing for steady flight at higher angles of attack.

A major difference appeared among the winglet families. In the location family, the front and mid winglets outperformed the back and baseline setups. This is because, as the winglet is closer to where the vortex forms, it gives a greater influence. The length family showed that the longer winglets generated more lift while reducing drag, while the small winglet remained close to the baseline results. For the angle family, the 20-degree and 40-degree wings gave the greatest lift values, while the 60-degree winglet was less effective and generated the most drag. These results help support the idea that winglet geometry plays a big role in shaping wing tip vortices.

For all families, the impact of different Reynolds numbers was clear. When at higher Reynolds numbers of about 300,000, the wing produced a higher lift and lower drag. This is because when at a higher Reynolds number, the boundary layer remains attached farther along the airfoil, resulting in improved lift and reduced drag.

Winglets played a minor role in stall behavior. While there was no major stall angle shown in the plots, the longer and more effective winglets maintained a higher lift value near stall since the winglets help reduce the wingtip vortices that cause drag. The reduction allowed the wing to have increased lift at higher angles of attack.

Finally, when comparing to thin airfoil theory, there was only slight agreement in the linear regions of the lift curve. The experimental slope was much lower than the Ideal thin airfoil theory slope. This is expected because thin airfoil theory assumes only 2-dimensional flow, no viscous effects, and no wingtip losses. The Eppler 212 wing is a finite wing with a small aspect ratio, so a good portion of lift is lost due to tip vortices, causing downwash. So, the experimental slope is very linear, following the same trend as the theoretical, just with a lower slope.

Experimental error may have had an impact on the results of the experiment. This can appear from small misalignments in the force balance setup, influencing the normal and axial forces data. Another source of error is that the data was measured throughout a long time period with a temperature change that may have affected the density values, which carries through to the aerodynamic coefficients. To reduce error in future experiments, one could perform more precise alignment of the force balance and do all trials at a similar time to avoid unique conditions.

The optimal winglet configuration for both lift and drag performance is the mid-mounted long winglet. It consistently produced strong lift and minimal drag at all angles of attack. The long winglet produced the largest lift increase and reduced the drag, especially at a higher Reynolds number. When mounted at the mid position, the winglet reduces the tip vortices before they can roll fully, which provides the greatest lift in the Location family. When looking at the angle family, it offered the greatest lift of all groups but came at the cost of much larger drag values when compared to the baseline test and other families. Depending on the mission of the aircraft, having an angled winglet can be useful if a greater lift is required at higher angles of attack. Nonetheless, for a balanced aerodynamic performance, having a vertical winglet is the best option to reduce drag. So the best engineering decision for the Eppler 212 airfoil families is the mid-mounted long winglet because it provides high lift gains without adding excessive drag.

In summary, the experiment was meaningful because it proved how changes in airfoil winglets will produce a major impact on aerodynamic behavior. The lab also proved key aerodynamic concepts such as downwash, induced drag, vortex formation at wing tips, and the effects of Reynolds number. The results show why winglets are a major piece in airfoil design and can be used to optimize performance for the desired purpose of the aircraft.

VI. **Appendix**

Appendix A: Tabulated Data

Table 1: Density and Uncertainty

Config	p_{atm}(kPa)	T(C°)	dp(kPa)	dT(C°)	Rho(kg/m^3)	drho(kg/m^3)
None	150.1	97.07	0.1	1	1.4127	0.0039301
20deg	150	97	0.1	1	1.412	0.0039291
40deg	150.5	96.7	0.1	1	1.4178	0.0039476
60deg	150	96.7	0.1	1	1.4131	0.0039353
Small	149.8	96.98	0.1	1	1.4102	0.0039245
Medium	150.1	96.99	0.1	1	1.413	0.0039317
Long	149.8	97	0.1	1	1.4101	0.0039241
Front	149.9	96.97	0.1	1	1.4112	0.0039272
Middle	150.4	96.8	0.1	1	1.4165	0.0039431
Back	149.9	96.7	0.1	1	1.4122	0.0039328

Table 2: C_L $vs. \alpha$ Plot Data

location'	front'	150000	-16.556332	-0.261601469	0.001412302
location'	front'	150000	-15.665448	-0.25650189	0.001405616
location'	front'	150000	-14.644807	-0.243768722	0.001389257
location'	front'	150000	-13.590665	-0.22282492	0.001364345
location'	front'	150000	-12.705974	-0.206653055	0.001346297
location'	front'	150000	-11.680438	-0.183375816	0.001322597
location'	front'	150000	-10.678989	-0.160804715	0.001302089
location'	front'	150000	-9.665254	-0.147496827	0.001291212
location'	front'	150000	-8.718593	-0.128146751	0.001276831
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location'	front'	150000	-6.709335	-0.082837663	0.001250714
location'	front'	150000	-5.61977	-0.050734068	0.001238815
location'	front'	150000	-4.713513	-0.03877379	0.001235825
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location'	front'	150000	-2.750681	0.007859495	0.001231823
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location'	front'	150000	3.29218	0.172576991	0.001312315
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angle'	60deg'	150000	-3.713501	0.019833012	0.001235795
angle'	60deg'	150000	-2.744507	0.049720065	0.001241617
angle'	60deg'	150000	-1.74837	0.075337813	0.0012505
angle'	60deg'	150000	-0.722337	0.106987719	0.001266379
angle'	60deg'	150000	0.339559	0.142012886	0.001289996
angle'	60deg'	150000	1.33423	0.178927423	0.001321437

angle'	60deg'	150000	2.245359	0.208587893	0.001351192
angle'	60deg'	150000	3.280378	0.241347928	0.001388393
angle'	60deg'	150000	4.315122	0.273906764	0.001429635
angle'	60deg'	150000	5.350391	0.30959885	0.001479232
angle'	60deg'	150000	6.293939	0.338898205	0.001523228
angle'	60deg'	150000	7.331331	0.367938564	0.00156929
angle'	60deg'	150000	8.306115	0.396149697	0.001616262
angle'	60deg'	150000	9.342961	0.420288871	0.001658299
angle'	60deg'	150000	11.224445	0.449773167	0.001711417
angle'	60deg'	150000	11.267217	0.454691264	0.001720567
angle'	60deg'	150000	12.278991	0.475275748	0.001759027
angle'	60deg'	150000	13.294784	0.488250359	0.001783717
angle'	60deg'	150000	14.354481	0.511087168	0.001827985
angle'	60deg'	150000	15.267356	0.526341994	0.001858078
angle'	60deg'	150000	16.289754	0.539934935	0.001885105
angle'	60deg'	150000	17.343895	0.556353079	0.001918684
angle'	60deg'	150000	18.267826	0.569309914	0.001945355
angle'	none'	150000	-16.735315	-0.260312398	0.001411169
angle'	none'	150000	-15.685475	-0.235265182	0.001380082
angle'	none'	150000	-14.680968	-0.206476697	0.001347133
angle'	none'	150000	-13.606854	-0.184704106	0.001324817
angle'	none'	150000	-12.726053	-0.171786125	0.001312708
angle'	none'	150000	-11.713418	-0.141434207	0.001287381
angle'	none'	150000	-10.698485	-0.115421698	0.001269266
angle'	none'	150000	-9.629359	-0.095901178	0.001258017
angle'	none'	150000	-8.734919	-0.082313676	0.001251289
angle'	none'	150000	-7.686032	-0.068559895	0.001245545
angle'	none'	150000	-6.675001	-0.042661634	0.00123753
angle'	none'	150000	-5.707898	-0.019019559	0.00123346
angle'	none'	150000	-4.722607	-0.003229198	0.001232467
angle'	none'	150000	-3.721322	0.018018206	0.001233322
angle'	none'	150000	-2.687209	0.041534325	0.00123725
angle'	none'	150000	-1.735096	0.05078501	0.001239638
angle'	none'	150000	-0.734275	0.061684341	0.001243023
angle'	none'	150000	0.305879	0.090271458	0.001255007
angle'	none'	150000	1.335779	0.117101396	0.00127028
angle'	none'	150000	2.265701	0.132808362	0.001280895
angle'	none'	150000	3.278877	0.155605485	0.001298409

angle'	none'	150000	4.27861	0.175603347	0.001315864
angle'	none'	150000	5.314618	0.20578449	0.001345713
angle'	none'	150000	6.331426	0.22989286	0.001372395
angle'	none'	150000	7.28645	0.249920408	0.001396361
angle'	none'	150000	8.303629	0.273976906	0.001427197
angle'	none'	150000	9.318792	0.301365841	0.00146492
angle'	none'	150000	10.282374	0.313403935	0.001482352
angle'	none'	150000	11.262425	0.338659366	0.001520379
angle'	none'	150000	12.303754	0.359040148	0.001552533
angle'	none'	150000	13.325557	0.381832272	0.001589852
angle'	none'	150000	14.345209	0.398556729	0.001618155
angle'	none'	150000	15.266473	0.424434909	0.001663134
angle'	none'	150000	16.292791	0.442206343	0.001695157
angle'	none'	150000	17.327575	0.459245716	0.001726536
angle'	none'	150000	17.34219	0.441105573	0.001693627
angle'	none'	150000	18.316324	0.470036758	0.001746791
angle'	20deg'	300000	-16.747521	-0.315824142	0.000383688
angle'	20deg'	300000	-15.670614	-0.285130077	0.00036906
angle'	20deg'	300000	-14.655842	-0.254272951	0.000355946
angle'	20deg'	300000	-13.610329	-0.226484376	0.000345028
angle'	20deg'	300000	-12.741557	-0.202248702	0.000337536
angle'	20deg'	300000	-11.708464	-0.170172317	0.000329117
angle'	20deg'	300000	-10.649535	-0.139491065	0.000322443
angle'	20deg'	300000	-9.54418	-0.101300861	0.000316034
angle'	20deg'	300000	-8.737141	-0.077225482	0.000312925
angle'	20deg'	300000	-7.680023	-0.042895953	0.000309967
angle'	20deg'	300000	-6.659905	-0.009472734	0.000308681
angle'	20deg'	300000	-5.697929	0.016841618	0.00030881
angle'	20deg'	300000	-4.722987	0.052321789	0.000310524
angle'	20deg'	300000	-3.718341	0.080060592	0.000313046
angle'	20deg'	300000	-2.718056	0.113008932	0.000317442
angle'	20deg'	300000	-1.609563	0.141777003	0.00032248
angle'	20deg'	300000	-0.73144	0.169816876	0.000328459
angle'	20deg'	300000	0.316966	0.201082929	0.000336259
angle'	20deg'	300000	1.344737	0.227275741	0.000343602
angle'	20deg'	300000	2.256636	0.249622345	0.000350592
angle'	20deg'	300000	3.308459	0.268865363	0.000356951
angle'	20deg'	300000	4.350877	0.299574107	0.000367827

angle'	20deg'	300000	5.342301	0.32184238	0.000376393
angle'	20deg'	300000	6.339856	0.346737831	0.000386543
angle'	20deg'	300000	7.271603	0.381422044	0.000401408
angle'	20deg'	300000	8.313782	0.40834658	0.00041385
angle'	20deg'	300000	9.193523	0.430162035	0.000424367
angle'	20deg'	300000	10.346719	0.454278387	0.000436311
angle'	20deg'	300000	11.252716	0.47194189	0.0004456
angle'	20deg'	300000	12.269835	0.490385049	0.000455691
angle'	20deg'	300000	13.282946	0.513222996	0.000468482
angle'	20deg'	300000	14.212231	0.529939532	0.000478185
angle'	20deg'	300000	15.380232	0.542559247	0.000486091
angle'	20deg'	300000	16.287197	0.559555899	0.000496084
angle'	20deg'	300000	17.355645	0.579397634	0.00050839
angle'	20deg'	300000	18.363939	0.595055902	0.000518559
angle'	40deg'	300000	-16.750371	-0.337637169	0.000394793
angle'	40deg'	300000	-15.694461	-0.311452317	0.000381654
angle'	40deg'	300000	-14.697516	-0.282283544	0.00036818
angle'	40deg'	300000	-13.739013	-0.249852679	0.000354382
angle'	40deg'	300000	-12.726567	-0.225989487	0.000345882
angle'	40deg'	300000	-11.703294	-0.197333824	0.000337442
angle'	40deg'	300000	-10.747881	-0.166140685	0.00032965
angle'	40deg'	300000	-9.756462	-0.136466041	0.000323378
angle'	40deg'	300000	-8.730795	-0.108182671	0.000318425
angle'	40deg'	300000	-7.677557	-0.073151136	0.000313913
angle'	40deg'	300000	-6.655688	-0.041787848	0.000311293
angle'	40deg'	300000	-5.740069	-0.009828331	0.000310102
angle'	40deg'	300000	-4.715657	0.02495646	0.000310461
angle'	40deg'	300000	-3.696108	0.054779885	0.000312103
angle'	40deg'	300000	-2.752954	0.085635303	0.000315146
angle'	40deg'	300000	-1.7483	0.119448034	0.000319975
angle'	40deg'	300000	-0.731261	0.145766981	0.000324751
angle'	40deg'	300000	0.312357	0.180693101	0.000332511
angle'	40deg'	300000	1.343349	0.211084533	0.000340442
angle'	40deg'	300000	2.26227	0.237841558	0.00034829
angle'	40deg'	300000	3.278053	0.264820694	0.00035696
angle'	40deg'	300000	4.306316	0.289202519	0.000365508
angle'	40deg'	300000	5.3448	0.319055239	0.000376735
angle'	40deg'	300000	6.247705	0.348125341	0.000388433

angle'	40deg'	300000	7.293836	0.377144034	0.000400929
angle'	40deg'	300000	8.315336	0.402129887	0.000412356
angle'	40deg'	300000	9.351448	0.42819784	0.000424826
angle'	40deg'	300000	10.270791	0.449149157	0.000435382
angle'	40deg'	300000	11.26235	0.467102528	0.000444748
angle'	40deg'	300000	12.294903	0.489767455	0.000457091
angle'	40deg'	300000	13.316706	0.506522784	0.000466507
angle'	40deg'	300000	14.330003	0.527368841	0.000478425
angle'	40deg'	300000	15.274244	0.543931759	0.000488143
angle'	40deg'	300000	16.300388	0.55905634	0.000497439
angle'	40deg'	300000	17.34563	0.576286379	0.000508476
angle'	40deg'	300000	18.25213	0.593030925	0.000519072
angle'	60deg'	300000	-16.726914	-0.341815117	0.000395587
angle'	60deg'	300000	-15.678337	-0.317608061	0.000383207
angle'	60deg'	300000	-14.662363	-0.289106949	0.000369663
angle'	60deg'	300000	-13.733909	-0.259583931	0.000356726
angle'	60deg'	300000	-12.710904	-0.232448116	0.000346949
angle'	60deg'	300000	-11.680939	-0.202852867	0.000338027
angle'	60deg'	300000	-10.726149	-0.175064975	0.000330737
angle'	60deg'	300000	-9.742823	-0.138779188	0.000322891
angle'	60deg'	300000	-8.724227	-0.109573481	0.000317679
angle'	60deg'	300000	-7.674177	-0.073705208	0.000313004
angle'	60deg'	300000	-6.673227	-0.042429857	0.000310362
angle'	60deg'	300000	-5.703574	-0.005509248	0.000309097
angle'	60deg'	300000	-4.71909	0.027749017	0.000309583
angle'	60deg'	300000	-3.713363	0.057518799	0.000311313
angle'	60deg'	300000	-2.646353	0.095992994	0.000315413
angle'	60deg'	300000	-1.682319	0.125995441	0.000320008
angle'	60deg'	300000	-0.718211	0.163061643	0.000327324
angle'	60deg'	300000	0.314594	0.199129946	0.000336073
angle'	60deg'	300000	1.344441	0.2246869	0.000343138
angle'	60deg'	300000	2.254437	0.253393646	0.00035199
angle'	60deg'	300000	3.383847	0.280461737	0.000361091
angle'	60deg'	300000	4.316893	0.306948311	0.000370762
angle'	60deg'	300000	5.329231	0.334755218	0.00038163
angle'	60deg'	300000	6.344162	0.369533451	0.000396151
angle'	60deg'	300000	7.303539	0.394101997	0.000407171
angle'	60deg'	300000	8.322582	0.418973598	0.00041885

angle'	60deg'	300000	9.32136	0.441093868	0.000429861
angle'	60deg'	300000	10.256236	0.45972839	0.000439451
angle'	60deg'	300000	11.268317	0.476553484	0.000448484
angle'	60deg'	300000	12.298433	0.496238487	0.000459354
angle'	60deg'	300000	13.330258	0.518583072	0.000471714
angle'	60deg'	300000	14.315747	0.53311309	0.000480381
angle'	60deg'	300000	15.284858	0.54844831	0.000489626
angle'	60deg'	300000	16.298156	0.561441077	0.000497967
angle'	60deg'	300000	17.3615	0.575070267	0.000506916
angle'	60deg'	300000	18.26266	0.589778205	0.000516545
angle'	none'	300000	-16.737938	-0.269288364	0.00036472
angle'	none'	300000	-15.695027	-0.237342234	0.000351807
angle'	none'	300000	-14.653915	-0.212201503	0.000341793
angle'	none'	300000	-13.759519	-0.190498233	0.00033444
angle'	none'	300000	-12.717904	-0.163270874	0.000327339
angle'	none'	300000	-11.689085	-0.135932936	0.00032154
angle'	none'	300000	-10.660254	-0.118698118	0.000318337
angle'	none'	300000	-9.662217	-0.096044889	0.00031491
angle'	none'	300000	-8.733564	-0.073389789	0.000312181
angle'	none'	300000	-7.68833	-0.047790191	0.000309939
angle'	none'	300000	-6.674968	-0.027394737	0.00030882
angle'	none'	300000	-5.733888	-0.007589124	0.000308316
angle'	none'	300000	-4.717975	0.013230748	0.000308394
angle'	none'	300000	-3.689013	0.03507	0.000309112
angle'	none'	300000	-2.651208	0.053109298	0.00031022
angle'	none'	300000	-1.705076	0.068184337	0.000311547
angle'	none'	300000	-0.729381	0.088182847	0.000313711
angle'	none'	300000	0.303275	0.110178318	0.00031679
angle'	none'	300000	1.342395	0.130010193	0.000320042
angle'	none'	300000	2.243649	0.147983876	0.000323552
angle'	none'	300000	3.270185	0.166281249	0.000327499
angle'	none'	300000	4.314937	0.189221924	0.000332976
angle'	none'	300000	5.334869	0.211687904	0.000338985
angle'	none'	300000	6.353264	0.234091147	0.000345696
angle'	none'	300000	7.280618	0.256452261	0.000352867
angle'	none'	300000	8.302083	0.278141023	0.000360318
angle'	none'	300000	9.335614	0.302112914	0.000369272
angle'	none'	300000	10.270094	0.324179315	0.000378022

angle'	none'	300000	11.255557	0.344836259	0.000386738
angle'	none'	300000	12.277636	0.363952034	0.000395315
angle'	none'	300000	13.315858	0.386042339	0.000405414
angle'	none'	300000	14.444096	0.412378044	0.000417961
angle'	none'	300000	15.257197	0.426930025	0.00042553
angle'	none'	300000	16.293084	0.446401776	0.000435712
angle'	none'	300000	17.294075	0.466576707	0.000446698
angle'	none'	300000	18.303377	0.485721739	0.00045756

Table 3: $C_L\ vs. C_L$ Plot Data

location'	front'	150000	-0.261601469	0.001412302	0.098618874	0.001278939
location'	front'	150000	-0.25650189	0.001405616	0.093998902	0.001275757
location'	front'	150000	-0.243768722	0.001389257	0.079397941	0.001267004
location'	front'	150000	-0.22282492	0.001364345	0.066683118	0.001259003
location'	front'	150000	-0.206653055	0.001346297	0.05391434	0.001252682
location'	front'	150000	-0.183375816	0.001322597	0.042509634	0.001246891
location'	front'	150000	-0.160804715	0.001302089	0.035481382	0.001243002
location'	front'	150000	-0.147496827	0.001291212	0.035454741	0.001241739
location'	front'	150000	-0.128146751	0.001276831	0.030221217	0.001239149
location'	front'	150000	-0.104220154	0.00126168	0.024450658	0.001236563
location'	front'	150000	-0.082837663	0.001250714	0.022970256	0.001235136
location'	front'	150000	-0.050734068	0.001238815	0.018490436	0.001233295
location'	front'	150000	-0.03877379	0.001235825	0.017002857	0.001232818
location'	front'	150000	-0.01458976	0.001232224	0.016430194	0.001232365
location'	front'	150000	0.007859495	0.001231823	0.018337527	0.001232504
location'	front'	150000	0.034280211	0.001234913	0.017180493	0.001232733
location'	front'	150000	0.057573822	0.001240868	0.018386229	0.001233513
location'	front'	150000	0.085339708	0.001251822	0.019301784	0.001234834
location'	front'	150000	0.110542976	0.001265326	0.020931429	0.001236537
location'	front'	150000	0.132504631	0.001279774	0.023175659	0.001238455
location'	front'	150000	0.172576991	0.001312315	0.030574664	0.001243308
location'	front'	150000	0.204410659	0.001343375	0.030146018	0.001246906
location'	front'	150000	0.226176553	0.001367214	0.033963168	0.001250438
location'	front'	150000	0.255617453	0.001402548	0.040164726	0.001256008
location'	front'	150000	0.286005674	0.001442411	0.044144291	0.001261902
location'	front'	150000	0.305390263	0.001469653	0.051257414	0.001267199
location'	front'	150000	0.338619293	0.00151907	0.058128004	0.00127564
location'	front'	150000	0.362368497	0.001556436	0.063184729	0.001282245

location'	front'	150000	0.390977408	0.001603655	0.073256265	0.00129229
location'	front'	150000	0.424401689	0.001661458	0.084640615	0.001305038
location'	front'	150000	0.448201606	0.001704296	0.09492242	0.001315915
location'	front'	150000	0.470656403	0.001746207	0.112441041	0.001331285
location'	front'	150000	0.480260816	0.001764204	0.114693536	0.001335213
location'	front'	150000	0.501823243	0.001805456	0.123901854	0.001346857
location'	front'	150000	0.524693496	0.001850245	0.135721776	0.001361249
location'	front'	150000	0.550339852	0.001901434	0.147313608	0.001377154
location'	mid'	150000	-0.293835968	0.001461631	0.097073436	0.001288958
location'	mid'	150000	-0.273382027	0.001433755	0.095474111	0.001284695
location'	mid'	150000	-0.25864283	0.001413914	0.079667346	0.001274859
location'	mid'	150000	-0.230003943	0.001378457	0.06165291	0.001263705
location'	mid'	150000	-0.215124323	0.001361388	0.052147102	0.00125873
location'	mid'	150000	-0.183163325	0.001328259	0.040749042	0.001251944
location'	mid'	150000	-0.164251691	0.001310778	0.030448707	0.001247907
location'	mid'	150000	-0.128118142	0.001282527	0.030776323	0.001244735
location'	mid'	150000	-0.115121012	0.001273942	0.030387039	0.001243702
location'	mid'	150000	-0.087923984	0.001258754	0.026706064	0.001241423
location'	mid'	150000	-0.059779406	0.001247224	0.025173592	0.001239925
location'	mid'	150000	-0.038556631	0.001241344	0.02069397	0.001238708
location'	mid'	150000	-0.014523524	0.001237785	0.02152621	0.001238415
location'	mid'	150000	0.004757195	0.001237225	0.018943448	0.001238064
location'	mid'	150000	0.033249739	0.001240305	0.022653079	0.001238829
location'	mid'	150000	0.057308991	0.001246365	0.021389773	0.001239343
location'	mid'	150000	0.079218582	0.001254697	0.023951506	0.001240586
location'	mid'	150000	0.106592949	0.001268739	0.027830692	0.001242707
location'	mid'	150000	0.134351331	0.001286843	0.024648884	0.00124429
location'	mid'	150000	0.158715242	0.001306	0.028246811	0.001247002
location'	mid'	150000	0.201810536	0.001346743	0.034674166	0.001252857
location'	mid'	150000	0.232539982	0.001380713	0.035969326	0.00125716
location'	mid'	150000	0.264953511	0.00142075	0.041178304	0.001263138
location'	mid'	150000	0.286166896	0.001449168	0.048012463	0.001268326
location'	mid'	150000	0.312204383	0.001486271	0.058403861	0.001276004
location'	mid'	150000	0.336495114	0.001522869	0.067060685	0.001283635
location'	mid'	150000	0.361093094	0.001561506	0.06847373	0.00128923
location'	mid'	150000	0.392345924	0.001613211	0.075883705	0.001299025
location'	mid'	150000	0.399651009	0.001625846	0.08247172	0.001303499
location'	mid'	150000	0.428933273	0.00167726	0.093852001	0.001315867

location'	mid'	150000	0.448804039	0.001713329	0.102630378	0.001325425
location'	mid'	150000	0.495232019	0.001800719	0.120685217	0.001348368
location'	mid'	150000	0.507647372	0.001824992	0.12929711	0.001357392
location'	mid'	150000	0.519313508	0.001848041	0.136968814	0.001365945
location'	mid'	150000	0.54825278	0.001905955	0.151146786	0.001384799
location'	mid'	150000	0.565614991	0.001941253	0.158171369	0.001395517
location'	back'	150000	-0.306428452	0.001475949	0.105042437	0.001292207
location'	back'	150000	-0.280503026	0.001438921	0.084667728	0.001277318
location'	back'	150000	-0.251971956	0.001401616	0.075236295	0.001268687
location'	back'	150000	-0.220691855	0.001363959	0.053507547	0.001256585
location'	back'	150000	-0.201320295	0.001342835	0.042959496	0.001251303
location'	back'	150000	-0.1784869	0.001320224	0.036680042	0.001247279
location'	back'	150000	-0.159946212	0.001303662	0.033640191	0.00124477
location'	back'	150000	-0.130309051	0.001280603	0.028614821	0.001241266
location'	back'	150000	-0.113834118	0.001269667	0.025288817	0.001239533
location'	back'	150000	-0.08250816	0.001252796	0.021952791	0.001237203
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location'	back'	150000	-0.036252369	0.00123757	0.021491119	0.001235455
location'	back'	150000	-0.012357868	0.001234288	0.018224088	0.001234734
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location'	back'	150000	0.052954531	0.001241754	0.024157307	0.001236254
location'	back'	150000	0.081636462	0.001252379	0.020146801	0.001236947
location'	back'	150000	0.108686532	0.001266654	0.03131208	0.00124013
location'	back'	150000	0.136039143	0.001284714	0.028993447	0.001241795
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location'	back'	150000	0.208838593	0.001350627	0.038296842	0.001251197
location'	back'	150000	0.232021602	0.001376584	0.042694066	0.001255283
location'	back'	150000	0.26286912	0.001414454	0.048007367	0.001261224
location'	back'	150000	0.299777156	0.001464454	0.056399781	0.00126986
location'	back'	150000	0.320659697	0.001494808	0.061219208	0.001275274
location'	back'	150000	0.345312717	0.001532421	0.067192788	0.001282232
location'	back'	150000	0.368021018	0.001568851	0.078183164	0.001291309
location'	back'	150000	0.386813684	0.001599827	0.080509479	0.001296468
location'	back'	150000	0.407060275	0.001634631	0.093153634	0.001306989
location'	back'	150000	0.424313277	0.00166514	0.104492023	0.001317044
location'	back'	150000	0.43042467	0.001676109	0.108261488	0.00132065

location'	back'	150000	0.440359849	0.001694007	0.112000869	0.001325288
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location'	front'	300000	-0.229821475	0.000349477	0.078099916	0.000370996
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location'	front'	300000	-0.166137664	0.000327898	0.033664729	0.000341029
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length'	med'	300000	0.426216125	0.000424382	0.094734644	0.000487062
length'	med'	300000	0.445936349	0.000434275	0.101992028	0.000500942
length'	med'	300000	0.458932697	0.0004415	0.109818923	0.000510502
length'	med'	300000	0.475353005	0.000450868	0.119832387	0.000522767
length'	med'	300000	0.488291557	0.000458348	0.127432676	0.000532532
length'	long'	300000	-0.255761355	0.000359826	0.094399895	0.000385056
length'	long'	300000	-0.239575439	0.00035285	0.083060042	0.000375898
length'	long'	300000	-0.216290454	0.000343651	0.066816043	0.000363553
length'	long'	300000	-0.193372507	0.000334863	0.04384579	0.000352038
length'	long'	300000	-0.170876027	0.000328813	0.035574846	0.00034265
length'	long'	300000	-0.156585764	0.000325314	0.029934654	0.000337175
length'	long'	300000	-0.126807923	0.000319363	0.024727852	0.00032732
length'	long'	300000	-0.10041503	0.00031515	0.021876222	0.000320178
length'	long'	300000	-0.077827847	0.000312315	0.020262045	0.000315308
length'	long'	300000	-0.059453611	0.000310485	0.018297817	0.000312195
length'	long'	300000	-0.033250614	0.000308725	0.016483343	0.000309174
length'	long'	300000	-0.005362785	0.000307907	0.014699166	0.000307806
length'	long'	300000	0.021469741	0.000308206	0.014651762	0.000308339
length'	long'	300000	0.040591334	0.000309046	0.015057493	0.00030981
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length'	long'	300000	0.340449289	0.000383495	0.059791758	0.000429454
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length'	long'	300000	0.473009767	0.00044732	0.111665064	0.000519966
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angle'	20deg'	150000	-0.306823881	0.00147468	0.10715905	0.001291834
angle'	20deg'	150000	-0.297423131	0.001460905	0.098540349	0.001285422
angle'	20deg'	150000	-0.273201499	0.001427243	0.083301703	0.001273859
angle'	20deg'	150000	-0.238910923	0.00138347	0.063685443	0.001260755
angle'	20deg'	150000	-0.200911158	0.001340728	0.046710771	0.001250552
angle'	20deg'	150000	-0.174742822	0.001315103	0.040511929	0.001246063
angle'	20deg'	150000	-0.151952673	0.00129533	0.033779052	0.001242401
angle'	20deg'	150000	-0.118155249	0.001270879	0.035153032	0.001239863
angle'	20deg'	150000	-0.091338014	0.001255492	0.030860851	0.001237349
angle'	20deg'	150000	-0.062460576	0.001243177	0.026040458	0.001235218
angle'	20deg'	150000	-0.029821063	0.001234799	0.025223706	0.001234172
angle'	20deg'	150000	-0.006484491	0.001232459	0.026636335	0.001234115
angle'	20deg'	150000	0.011937095	0.001232702	0.024273841	0.001233811
angle'	20deg'	150000	0.03975096	0.001236684	0.022594003	0.001234035
angle'	20deg'	150000	0.06755143	0.001244994	0.025233899	0.001235307
angle'	20deg'	150000	0.097138881	0.001258408	0.026437988	0.001236981
angle'	20deg'	150000	0.12028541	0.001272152	0.029944005	0.001239078
angle'	20deg'	150000	0.145187745	0.001289931	0.031211917	0.001241323
angle'	20deg'	150000	0.163338863	0.001304791	0.031664989	0.001243118
angle'	20deg'	150000	0.201912868	0.001341483	0.034559064	0.001247955
angle'	20deg'	150000	0.224968655	0.001366547	0.037255557	0.001251487

angle'	20deg'	150000	0.267549673	0.001418589	0.047072782	0.001260114
angle'	20deg'	150000	0.282468533	0.00143833	0.046111528	0.001262347
angle'	20deg'	150000	0.317230016	0.001487682	0.058143486	0.001272017
angle'	20deg'	150000	0.338916809	0.001520358	0.062183273	0.001277579
angle'	20deg'	150000	0.359684613	0.001553057	0.068799407	0.00128422
angle'	20deg'	150000	0.395293065	0.001611906	0.079946796	0.001296574
angle'	20deg'	150000	0.424144888	0.001661886	0.087271103	0.001306727
angle'	20deg'	150000	0.452094634	0.001712149	0.094921181	0.001317489
angle'	20deg'	150000	0.4641714	0.001734813	0.107743593	0.001327436
angle'	20deg'	150000	0.481560374	0.001767636	0.118183505	0.001338223
angle'	20deg'	150000	0.50470937	0.001812375	0.134100907	0.00135494
angle'	20deg'	150000	0.525197238	0.00185228	0.13909026	0.001364298
angle'	20deg'	150000	0.529632237	0.001861331	0.145633915	0.00137029
angle'	20deg'	150000	0.552877754	0.001907839	0.154973369	0.00138426
angle'	20deg'	150000	0.574220806	0.00195159	0.168461346	0.001401612
angle'	40deg'	150000	-0.352037686	0.001551323	0.116321102	0.001312575
angle'	40deg'	150000	-0.321905582	0.001504234	0.100270327	0.001297386
angle'	40deg'	150000	-0.290458708	0.00145815	0.081584431	0.001282541
angle'	40deg'	150000	-0.259949692	0.001416891	0.065186618	0.001270978
angle'	40deg'	150000	-0.226869687	0.0013764	0.052967275	0.001262173
angle'	40deg'	150000	-0.206207284	0.001353279	0.040653464	0.001256283
angle'	40deg'	150000	-0.176004693	0.001323135	0.033774249	0.001251357
angle'	40deg'	150000	-0.144824723	0.001296537	0.031647222	0.001247922
angle'	40deg'	150000	-0.120791314	0.001279215	0.026728272	0.001245171
angle'	40deg'	150000	-0.091776883	0.001262298	0.024139613	0.001242915
angle'	40deg'	150000	-0.062167932	0.001249704	0.025832049	0.001241755
angle'	40deg'	150000	-0.030351201	0.001241415	0.020930155	0.001240209
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angle'	40deg'	150000	0.263240482	0.001420342	0.037202486	0.001263633
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angle'	40deg'	150000	0.326576423	0.001509426	0.054213408	0.001279075
angle'	40deg'	150000	0.359525546	0.001560818	0.060593343	0.001287766
angle'	40deg'	150000	0.388496124	0.001608582	0.067437564	0.001296511
angle'	40deg'	150000	0.4163778	0.001656758	0.078534374	0.0013074
angle'	40deg'	150000	0.437845077	0.001695116	0.086734086	0.001316315
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angle'	40deg'	150000	0.584073763	0.00198161	0.161181524	0.001405294
angle'	60deg'	150000	-0.34994843	0.001543319	0.12321319	0.00131238
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angle'	60deg'	150000	-0.299930314	0.001466918	0.08987987	0.001283944
angle'	60deg'	150000	-0.273872771	0.001430549	0.073134856	0.001272111
angle'	60deg'	150000	-0.242855335	0.001390705	0.05608807	0.001261286
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angle'	none'	150000	0.424434909	0.001663134	0.096882642	0.001311694
angle'	none'	150000	0.442206343	0.001695157	0.106667799	0.001321356
angle'	none'	150000	0.459245716	0.001726536	0.116252558	0.00133131
angle'	none'	150000	0.441105573	0.001693627	0.115573102	0.001326222
angle'	none'	150000	0.470036758	0.001746791	0.123478453	0.001338637
angle'	20deg'	300000	-0.315824142	0.000383688	0.111308418	0.000419728
angle'	20deg'	300000	-0.285130077	0.00036906	0.091367925	0.000400482
angle'	20deg'	300000	-0.254272951	0.000355946	0.07254529	0.000382601
angle'	20deg'	300000	-0.226484376	0.000345028	0.052518945	0.000367604
angle'	20deg'	300000	-0.202248702	0.000337536	0.043716569	0.000356168
angle'	20deg'	300000	-0.170172317	0.000329117	0.03587905	0.000342768
angle'	20deg'	300000	-0.139491065	0.000322443	0.030716059	0.000331823
angle'	20deg'	300000	-0.101300861	0.000316034	0.027630566	0.000320977
angle'	20deg'	300000	-0.077225482	0.000312925	0.024962332	0.000315741
angle'	20deg'	300000	-0.042895953	0.000309967	0.023062906	0.000310665
angle'	20deg'	300000	-0.009472734	0.000308681	0.021815484	0.000308473
angle'	20deg'	300000	0.016841618	0.00030881	0.021697026	0.000308709
angle'	20deg'	300000	0.052321789	0.000310524	0.02189317	0.000311727
angle'	20deg'	300000	0.080060592	0.000313046	0.021681149	0.000316174
angle'	20deg'	300000	0.113008932	0.000317442	0.022746364	0.000323776
angle'	20deg'	300000	0.141777003	0.00032248	0.02472166	0.000332345
angle'	20deg'	300000	0.169816876	0.000328459	0.028069926	0.00034232
angle'	20deg'	300000	0.201082929	0.000336259	0.032349409	0.000355141
angle'	20deg'	300000	0.227275741	0.000343602	0.035574119	0.000367102
angle'	20deg'	300000	0.249622345	0.000350592	0.040238125	0.000378196
angle'	20deg'	300000	0.268865363	0.000356951	0.043499148	0.000388262
angle'	20deg'	300000	0.299574107	0.000367827	0.048837925	0.000405282
angle'	20deg'	300000	0.32184238	0.000376393	0.054232196	0.000418366
angle'	20deg'	300000	0.346737831	0.000386543	0.060905678	0.000433632
angle'	20deg'	300000	0.381422044	0.000401408	0.069476466	0.000455796
angle'	20deg'	300000	0.40834658	0.00041385	0.07834465	0.000473822
angle'	20deg'	300000	0.430162035	0.000424367	0.08592788	0.00048885
angle'	20deg'	300000	0.454278387	0.000436311	0.093999396	0.000505808

angle'	20deg'	300000	0.47194189	0.0004456	0.101755414	0.000518608
angle'	20deg'	300000	0.490385049	0.000455691	0.110787114	0.000532255
angle'	20deg'	300000	0.513222996	0.000468482	0.121865209	0.000549417
angle'	20deg'	300000	0.529939532	0.000478185	0.130632538	0.00056222
angle'	20deg'	300000	0.542559247	0.000486091	0.139401163	0.000572203
angle'	20deg'	300000	0.559555899	0.000496084	0.147447597	0.000585384
angle'	20deg'	300000	0.579397634	0.00050839	0.158778175	0.000601146
angle'	20deg'	300000	0.595055902	0.000518559	0.168929143	0.00061385
angle'	40deg'	300000	-0.337637169	0.000394793	0.116915234	0.000434344
angle'	40deg'	300000	-0.311452317	0.000381654	0.099874878	0.00041726
angle'	40deg'	300000	-0.282283544	0.00036818	0.081200725	0.000399308
angle'	40deg'	300000	-0.249852679	0.000354382	0.057893465	0.000380651
angle'	40deg'	300000	-0.225989487	0.000345882	0.044910886	0.000368331
angle'	40deg'	300000	-0.197333824	0.000337442	0.036991305	0.000355173
angle'	40deg'	300000	-0.166140685	0.00032965	0.031844021	0.000342578
angle'	40deg'	300000	-0.136466041	0.000323378	0.027601181	0.000332282
angle'	40deg'	300000	-0.108182671	0.000318425	0.023177568	0.000324106
angle'	40deg'	300000	-0.073151136	0.000313913	0.020404512	0.000316472
angle'	40deg'	300000	-0.041787848	0.000311293	0.017966746	0.000312039
angle'	40deg'	300000	-0.009828331	0.000310102	0.017123616	0.000309999
angle'	40deg'	300000	0.02495646	0.000310461	0.016874705	0.000310638
angle'	40deg'	300000	0.054779885	0.000312103	0.01643511	0.00031353
angle'	40deg'	300000	0.085635303	0.000315146	0.017416215	0.000318771
angle'	40deg'	300000	0.119448034	0.000319975	0.019305644	0.000326995
angle'	40deg'	300000	0.145766981	0.000324751	0.020781051	0.000335063
angle'	40deg'	300000	0.180693101	0.000332511	0.024918104	0.000347897
angle'	40deg'	300000	0.211084533	0.000340442	0.02886519	0.000360822
angle'	40deg'	300000	0.237841558	0.00034829	0.032782322	0.000373423
angle'	40deg'	300000	0.264820694	0.00035696	0.036920408	0.000387166
angle'	40deg'	300000	0.289202519	0.000365508	0.041973394	0.000400447
angle'	40deg'	300000	0.319055239	0.000376735	0.048373902	0.000417657
angle'	40deg'	300000	0.348125341	0.000388433	0.054983028	0.000435321
angle'	40deg'	300000	0.377144034	0.000400929	0.062945678	0.000453807
angle'	40deg'	300000	0.402129887	0.000412356	0.071075213	0.000470364
angle'	40deg'	300000	0.42819784	0.000424826	0.080036575	0.000488174
angle'	40deg'	300000	0.449149157	0.000435382	0.088527358	0.000502926
angle'	40deg'	300000	0.467102528	0.000444748	0.096291738	0.000515835
angle'	40deg'	300000	0.489767455	0.000457091	0.107288086	0.000532517

angle'	40deg'	300000	0.506522784	0.000466507	0.115766486	0.000545077
angle'	40deg'	300000	0.527368841	0.000478425	0.126046932	0.000560895
angle'	40deg'	300000	0.543931759	0.000488143	0.134511552	0.00057365
angle'	40deg'	300000	0.55905634	0.000497439	0.143536272	0.000585551
angle'	40deg'	300000	0.576286379	0.000508476	0.154968986	0.000599375
angle'	40deg'	300000	0.593030925	0.000519072	0.164732233	0.000612795
angle'	60deg'	300000	-0.341815117	0.000395587	0.118959236	0.000436291
angle'	60deg'	300000	-0.317608061	0.000383207	0.102670417	0.000420279
angle'	60deg'	300000	-0.289106949	0.000369663	0.083405792	0.00040239
angle'	60deg'	300000	-0.259583931	0.000356726	0.061653387	0.000384992
angle'	60deg'	300000	-0.232448116	0.000346949	0.048349685	0.000370701
angle'	60deg'	300000	-0.202852867	0.000338027	0.040218782	0.000356788
angle'	60deg'	300000	-0.175064975	0.000330737	0.034208483	0.000345119
angle'	60deg'	300000	-0.138779188	0.000322891	0.029888071	0.000332137
angle'	60deg'	300000	-0.109573481	0.000317679	0.024943621	0.000323534
angle'	60deg'	300000	-0.073705208	0.000313004	0.021847467	0.000315603
angle'	60deg'	300000	-0.042429857	0.000310362	0.019341082	0.000311118
angle'	60deg'	300000	-0.005509248	0.000309097	0.018678225	0.000308927
angle'	60deg'	300000	0.027749017	0.000309583	0.018002642	0.00030982
angle'	60deg'	300000	0.057518799	0.000311313	0.017198227	0.000312905
angle'	60deg'	300000	0.095992994	0.000315413	0.017914172	0.000320028
angle'	60deg'	300000	0.125995441	0.000320008	0.019465381	0.000327895
angle'	60deg'	300000	0.163061643	0.000327324	0.022856439	0.000340201
angle'	60deg'	300000	0.199129946	0.000336073	0.027143661	0.000354651
angle'	60deg'	300000	0.2246869	0.000343138	0.029941614	0.00036619
angle'	60deg'	300000	0.253393646	0.00035199	0.034563073	0.000380362
angle'	60deg'	300000	0.280461737	0.000361091	0.039023102	0.000394743
angle'	60deg'	300000	0.306948311	0.000370762	0.04467123	0.000409725
angle'	60deg'	300000	0.334755218	0.00038163	0.051134636	0.000426299
angle'	60deg'	300000	0.369533451	0.000396151	0.059644688	0.000448097
angle'	60deg'	300000	0.394101997	0.000407171	0.06741349	0.000464225
angle'	60deg'	300000	0.418973598	0.00041885	0.075802681	0.000481065
angle'	60deg'	300000	0.441093868	0.000429861	0.085028951	0.000496541
angle'	60deg'	300000	0.45972839	0.000439451	0.093051368	0.000509856
angle'	60deg'	300000	0.476553484	0.000448484	0.101284589	0.00052215
angle'	60deg'	300000	0.496238487	0.000459354	0.111172027	0.000536778
angle'	60deg'	300000	0.518583072	0.000471714	0.120887387	0.00055351
angle'	60deg'	300000	0.53311309	0.000480381	0.129553316	0.000564754

angle'	60deg'	300000	0.54844831	0.000489626	0.138403373	0.000576717
angle'	60deg'	300000	0.561441077	0.000497967	0.147636171	0.000587132
angle'	60deg'	300000	0.575070267	0.000506916	0.157548078	0.000598186
angle'	60deg'	300000	0.589778205	0.000516545	0.167476932	0.000610137
angle'	none'	300000	-0.269288364	0.00036472	0.095179325	0.000392483
angle'	none'	300000	-0.237342234	0.000351807	0.077805914	0.000374733
angle'	none'	300000	-0.212201503	0.000341793	0.057606326	0.000361443
angle'	none'	300000	-0.190498233	0.00033444	0.041553744	0.000351141
angle'	none'	300000	-0.163270874	0.000327339	0.031599709	0.000340076
angle'	none'	300000	-0.135932936	0.00032154	0.02605379	0.000330582
angle'	none'	300000	-0.118698118	0.000318337	0.0216118	0.000325348
angle'	none'	300000	-0.096044889	0.00031491	0.019126224	0.000319536
angle'	none'	300000	-0.073389789	0.000312181	0.016948165	0.000314875
angle'	none'	300000	-0.047790191	0.000309939	0.014423436	0.000311046
angle'	none'	300000	-0.027394737	0.00030882	0.012847983	0.000309134
angle'	none'	300000	-0.007589124	0.000308316	0.012177825	0.000308267
angle'	none'	300000	0.013230748	0.000308394	0.012028873	0.00030841
angle'	none'	300000	0.03507	0.000309112	0.011459155	0.0003097
angle'	none'	300000	0.053109298	0.00031022	0.011563345	0.000311651
angle'	none'	300000	0.068184337	0.000311547	0.013559215	0.000313912
angle'	none'	300000	0.088182847	0.000313711	0.013920071	0.000317689
angle'	none'	300000	0.110178318	0.00031679	0.016677671	0.000322932
angle'	none'	300000	0.130010193	0.000320042	0.017494217	0.000328519
angle'	none'	300000	0.147983876	0.000323552	0.021212905	0.000334353
angle'	none'	300000	0.166281249	0.000327499	0.023990042	0.000340918
angle'	none'	300000	0.189221924	0.000332976	0.026792848	0.000349996
angle'	none'	300000	0.211687904	0.000338985	0.030524044	0.000359789
angle'	none'	300000	0.234091147	0.000345696	0.036365529	0.000370434
angle'	none'	300000	0.256452261	0.000352867	0.041274697	0.000381758
angle'	none'	300000	0.278141023	0.000360318	0.046160301	0.000393386
angle'	none'	300000	0.302112914	0.000369272	0.053389384	0.000407008
angle'	none'	300000	0.324179315	0.000378022	0.060236393	0.000420136
angle'	none'	300000	0.344836259	0.000386738	0.067625317	0.00043295
angle'	none'	300000	0.363952034	0.000395315	0.07567066	0.000445264
angle'	none'	300000	0.386042339	0.000405414	0.083572924	0.000459802
angle'	none'	300000	0.412378044	0.000417961	0.093117491	0.000477657
angle'	none'	300000	0.426930025	0.00042553	0.100764178	0.00048795
angle'	none'	300000	0.446401776	0.000435712	0.109824852	0.000501863

angle'	none'	300000	0.466576707	0.000446698	0.119937129	0.000516617
angle'	none'	300000	0.485721739	0.00045756	0.130314468	0.000530938

Appendix B: Sample Calculations

$F_{A,\text{corr}} = F_{A,\text{meas}} - F_{A,\text{grav}}$
$F_{N,\text{corr}} = F_{N,\text{meas}} - F_{N,\text{grav}}$

Givens:
Measured:
$F_{N,\text{meas}} = 1.82\ \text{N}$
$F_{A,\text{meas}} = 0.41\ \text{N}$
Gravity fit data:
$F_{N,\text{grav}}(10°) = 0.14\ \text{N}$
$F_{A,\text{grav}}(10°) = 0.03\ \text{N}$
Corrected forces (using Eq. (1)-(2)):
$F_{A,\text{corr}} = F_{A,\text{meas}} - F_{A,\text{grav}}$
$F_{A,\text{corr}} = 0.41 - 0.03 = 0.38\ \text{N}$
$F_{N,\text{corr}} = F_{N,\text{meas}} - F_{N,\text{grav}}$
$F_{N,\text{corr}} = 1.82 - 0.14 = 1.68\ \text{N}$

Freestream Density:
Density is computed with the ideal-gas law, Eq. (4):
$$\rho = \frac{p}{R\,T}$$

Given:
$p = 97.0\ \text{kPa} = 97000\ \text{Pa}$
$T_C = 26°\text{C} \Rightarrow T = 299.15\ \text{K}$
$R = 287\ \text{J/kg*K}$
$$\rho = \frac{97000}{287 \times 299.15} = 1.13\ \text{kg/m}^3$$

Dynamic Viscosity (Sutherland's Law) Using Eq. (5):
$$\mu = \mu_{\text{ref}}(\frac{T}{T_{\text{ref}}})^{3/2}\frac{T_{\text{ref}} + S_\mu}{T + S_\mu}$$

Given:
$\mu_{\text{ref}} = 1.716 \times 10^{-5}$ (kg/m*s)
$T_{\text{ref}} = 273.15\ \text{K}$
$S_\mu = 110.4\ \text{K}$

$$\mu = 1.716 \times 10^{-5}(\frac{299.15}{273.15})^{3/2}\frac{273.15 + 110.4}{299.15 + 110.4} = 1.85 \times 10^{-5}\ \text{(kg/m*s)}$$

Velocity is found from the Reynolds number
$$V_\infty = \frac{Re\mu}{\rho c}$$

Given:
$Re = 300{,}000$
$c = 0.1397$ m
$$V_\infty = \frac{3.0 \times 10^5 (1.85 \times 10^{-5})}{1.13(0.1397)} = 35.3 \text{ m/s}$$

Dynamic pressure Eq. (3):
$$q = \frac{1}{2}\rho V_\infty^2 = \frac{1}{2}(1.13)(35.3^2) = 704 \text{ Pa}$$

Lift and Drag Calculation Eqs. (6)–(7):
$L = F_{N,\mathrm{corr}} \cos\alpha - F_{A,\mathrm{corr}} \sin\alpha$
$D = F_{A,\mathrm{corr}} \cos\alpha + F_{N,\mathrm{corr}} \sin\alpha$
At $\alpha = 10°$:
$L = 1.68\cos(10°) - 0.38\sin(10°) = 1.52$ N
$D = 0.38\cos(10°) + 1.68\sin(10°) = 0.68$ N

Coefficients C_L and C_D
Using Eqs. (7)–(8):
$$C_L = \frac{L}{qS} C_D = \frac{D}{qS}$$

Given:
$S = c\, b = 0.1397 \times 0.1524 = 0.02129$ m^2:
$$C_L = \frac{1.52}{704(0.02129)} = 0.100$$
$$C_D = \frac{0.68}{704(0.02129)} = 0.045$$

Sample Uncertainty Propagation
Using Eqs. (9)–(10):
$$\delta C_L = \sqrt{\left(\frac{\partial C_L}{\partial F_N}\,\delta F_N\right)^2 + \left(\frac{\partial C_L}{\partial F_A}\,\delta F_A\right)^2 + \left(\frac{\partial C_L}{\partial q}\,\delta q\right)^2 + \left(\frac{\partial C_L}{\partial \alpha}\,\delta \alpha\right)^2}$$
$$\delta C_D = \sqrt{(\frac{\partial C_D}{\partial F_N}\,\delta F_N)^2 + (\frac{\partial C_D}{\partial F_A}\,\delta F_A)^2 + (\frac{\partial C_D}{\partial q}\,\delta q)^2 + (\frac{\partial C_D}{\partial \alpha}\,\delta \alpha)^2}$$

Using uncertainty values
$\delta F_N = \delta F_A = 0.005$ N,
$\delta q = 0.5$ Pa,
$\delta\alpha = 0.05°$
Final sample uncertainty:
$\delta C_L \approx 0.004$
$\delta C_D \approx 0.003$

Appendix C: MATLAB Code

```
function AeroLab4_G7()
clc; close all;
%% AEE 360 – Lab 4 (Group 7)

%% Files
ALL_FILES = { ...
   'G7_Conditions.txt', ...
   'Lab4_G7_20deg_150k.csv', 'Lab4_G7_20deg_300k.csv', 'Lab4_G7_20deg_grav.csv', ...
   'Lab4_G7_40deg_150k.csv', 'Lab4_G7_40deg_300k.csv', 'Lab4_G7_40deg_grav.csv', ...
   'Lab4_G7_60deg_150k.csv', 'Lab4_G7_60deg_300k.csv', 'Lab4_G7_60deg_grav.csv', ...
   'Lab4_G7_back_150k.csv',  'Lab4_G7_back_300k.csv',  'Lab4_G7_back_grav.csv', ...
   'Lab4_G7_front_150k.csv', 'Lab4_G7_front_300k.csv', 'Lab4_G7_front_grav.csv', ...
   'Lab4_G7_long_150k.csv',  'Lab4_G7_long_300k.csv',  'Lab4_G7_long_grav.csv', ...
   'Lab4_G7_med_150k.csv',   'Lab4_G7_med_300k.csv',   'Lab4_G7_med_grav.csv', ...
   'Lab4_G7_mid_150k.csv',   'Lab4_G7_mid_300k.csv',   'Lab4_G7_mid_grav.csv', ...
   'Lab4_G7_none_150k.csv',  'Lab4_G7_none_300k.csv',  'Lab4_G7_none_grav.csv', ...
   'Lab4_G7_small_150k.csv', 'Lab4_G7_small_300k.csv', 'Lab4_G7_small_grav.csv' ...
};

%% Geometry, outputs, constants
c = 0.1397; % chord [m]
b = 0.1524; % span [m]
S = c*b; % area [m^2]

odir_fig = fullfile(pwd,'figs');
if ~exist(odir_fig,'dir'), mkdir(odir_fig);
end
odir_out = fullfile(pwd,'out');
if ~exist(odir_out,'dir'), mkdir(odir_out);
end

% Uncertainties
dFN = 0.005; % N
dFA = 0.005; % N
dAlpha_deg = 0.05; % deg
dq_abs = 0.5; % Pa

% Re
Re_list = [150e3, 300e3];

% Families
FAMILIES.location = {'front','mid','back','none'};
FAMILIES.length = {'small','med','long','none'};
FAMILIES.angle = {'20deg','40deg','60deg','none'};

deg2rad_local = @(d) d.*(pi/180);
san_vars = @(V) cellfun(@(x)regexprep(lower(x),'[^a-z0-9]',''), V, 'uni',0);
```

```
%% index of data files
IDX = containers.Map('KeyType','char','ValueType','any');
condFile = '';
for i = 1:numel(ALL_FILES)
    f = ALL_FILES{i};
    if strcmpi(f,'G7_Conditions.txt')
        condFile = f;
        continue;
    end
    if isempty(regexpi(f,'^Lab4_G7_','once')), continue; end
    tok = regexp(lower(f), '^lab4_g7_([a-z0-9]+)_([a-z0-9]+)\.csv$', 'tokens','once');
    if isempty(tok), continue; end
    tag = tok{1}; % e.g., 'front','mid','20deg','none',...
    tail = tok{2}; % '150k' '300k' 'grav'
    if ~isKey(IDX, tag)
        IDX(tag) = struct('f150k','','f300k','','fgrav','');
    end
    e = IDX(tag);
    switch tail
        case '150k', e.f150k = f;
        case '300k', e.f300k = f;
        case 'grav', e.fgrav = f;
    end
    IDX(tag) = e;
end

%% G7_Conditions
COND = containers.Map('KeyType','char','ValueType','any');

if ~isempty(condFile) && isfile(condFile)
    optsC = detectImportOptions(condFile,'NumHeaderLines',0);
    optsC.VariableNamingRule = 'preserve';
    Tcond = readtable(condFile,optsC);

    namesC = san_vars(Tcond.Properties.VariableNames);
    iCfg = find(contains(namesC,{'wingletconfig','config','winglet'}),1,'first');
    iT = find(contains(namesC,{'temp','temperature'}),1,'first');
    iP = find(contains(namesC,{'pressure','pkpa'}),1,'first');

    if isempty(iCfg) || isempty(iT) || isempty(iP)
        warning('G7_Conditions.txt fail');
    else
        cfgCol = Tcond{:,iCfg};
        Tcol = Tcond{:,iT};
        Pcol = Tcond{:,iP};

        for r = 1:height(Tcond)
```

```
            % winglet config
            cfgRaw = lower(strtrim(string(cfgCol(r))));
            cfgRaw = char(cfgRaw);

            switch cfgRaw
                case {'none','no winglet'}
                    tag = 'none';
                case {'20deg','20 deg','20°'}
                    tag = '20deg';
                case {'40deg','40 deg','40°'}
                    tag = '40deg';
                case {'60deg','60 deg','60°'}
                    tag = '60deg';
                case {'front'}
                    tag = 'front';
                case {'middle','mid'}
                    tag = 'mid';
                case {'back'}
                    tag = 'back';
                case {'small'}
                    tag = 'small';
                case {'medium','med'}
                    tag = 'med';
                case {'long'}
                    tag = 'long';
                otherwise
                    tag = cfgRaw;
            end

            amb.TC = Tcol(r); % °C
            amb.PkPa = Pcol(r); % kPa
            COND(tag) = amb;
        end
    end
end

%% Process
SETS = struct();
families = fieldnames(FAMILIES);

for fi = 1:numel(families)
    fam = families{fi};
    fprintf('Family: %s\n', upper(fam));
    for r = 1:numel(Re_list)
        Re = Re_list(r);
        curves = [];
        tags = FAMILIES.(fam);
```

```
for k = 1:numel(tags)
    tag = tags{k};
    if ~isKey(IDX, tag), continue; end
    entry = IDX(tag);

    % file by Re
    if Re < 2e5
        ftest = entry.f150k;
    else
        ftest = entry.f300k;
    end
    fgrav = entry.fgrav;
    if isempty(ftest) || ~isfile(ftest), continue;
    end

    % default ambient
    PkPa = 97.0;
    TC = 26.0;
    if isKey(COND, tag)
        amb = COND(tag);
        PkPa = amb.PkPa;
        TC = amb.TC;
    end

    % --- read test table
    opts = detectImportOptions(ftest, 'NumHeaderLines',0);
    opts.VariableNamingRule = 'preserve';
    Ttbl = readtable(ftest, opts);
    names = san_vars(Ttbl.Properties.VariableNames);
    pick = @(keys) find(contains(names,keys,'IgnoreCase',true),1,'first');
    ia = pick({'alpha','pitchangle','angle','deg'});
    iN = pick({'pgbnormal','fn','normal'});
    iA = pick({'pgbaxial','fa','axial'});
    if isempty(ia) || isempty(iN) || isempty(iA)
        nmask = varfun(@isnumeric,Ttbl,'OutputFormat','uniform');
        idx = find(nmask);
        if numel(idx) >= 4
            ia = idx(end);
            iN = idx(end-2);
            iA = idx(end-3);
        else
            warning('Could not parse alpha/FN/FA from %s', ftest);
            continue;
        end
    end
    alpha = Ttbl{:,ia};
    FN = Ttbl{:,iN};
    FA = Ttbl{:,iA};
```

```
% drop any NaN rows in test data
maskT = isfinite(alpha) & isfinite(FN) & isfinite(FA);
alpha = alpha(maskT);
FN = FN(maskT);
FA = FA(maskT);
if numel(alpha) < 3
    warning('Not enough valid data for tag %s Re=%g; skipping.', tag, Re);
    continue;
end

% Gravity correction
if ~isempty(fgrav) && isfile(fgrav)
    gopts = detectImportOptions(fgrav,'NumHeaderLines',0);
    gopts.VariableNamingRule = 'preserve';
    Gtbl = readtable(fgrav,gopts);
    gnames = san_vars(Gtbl.Properties.VariableNames);
    ga = find(contains(gnames,{'alpha','pitchangle','angle','deg'}),1,'first');
    gN = find(contains(gnames,{'pgbnormal','fn','normal'}),1,'first');
    gA = find(contains(gnames,{'pgbaxial','fa','axial'}),1,'first');
    FNc = FN;
    FAc = FA;
    if ~(isempty(ga) || isempty(gN) || isempty(gA))
        Ga = Gtbl{:,ga};
        GN = Gtbl{:,gN};
        GA = Gtbl{:,gA};
        maskG = isfinite(Ga) & isfinite(GN) & isfinite(GA);
        Ga = Ga(maskG); GN = GN(maskG); GA = GA(maskG);
        if numel(Ga) >= 3
            pFN = polyfit(Ga, GN, 2);
            pFA = polyfit(Ga, GA, 2);
            FNc = FN - polyval(pFN, alpha);
            FAc = FA - polyval(pFA, alpha);
        else
            warning('Not enough clean grav points for %s; skipping grav correction.', tag);
        end
    else
        warning('Could not parse gravity file %s; skipping grav correction.', fgrav);
    end
else
    FNc = FN;
    FAc = FA;
end

% Properties & q
P = PkPa*1000;
T = TC + 273.15;
R = 287.05;
```

```
rho = P/(R*T);
% Sutherland
Tref = 273.15;
muref = 1.716e-5;
Suth = 110.4;
mu = muref * (T/Tref)^(3/2) * (Tref+Suth)/(T+Suth);

V = Re * mu / (rho * c); % Re = rho V c / mu
q = 0.5 * rho * V^2; % Pa
dq = dq_abs; % Pa

ca = cosd(alpha);
sa = sind(alpha);
Ltest = FNc.*ca - FAc.*sa;
Dtest = FNc.*sa + FAc.*ca;

% Aero coefficients
L = Ltest;
D = Dtest;
CL = L./(q*S);
CD = D./(q*S);

% Uncertainty propagation
dA = deg2rad_local(dAlpha_deg);
dCL_dFN = ca./(q*S);
dCL_dFA = -sa./(q*S);
dCL_dq = -CL./q;
dCD_dFN = sa./(q*S);
dCD_dFA = ca./(q*S);
dCD_dq = -CD./q;
dCL_dA = -CD;
dCD_dA =  CL;
% combine
dCL = sqrt( (dCL_dFN*dFN).^2 + (dCL_dFA*dFA).^2 + (dCL_dA*dA).^2 + (dCL_dq*dq).^2 );
dCD = sqrt( (dCD_dFN*dFN).^2 + (dCD_dFA*dFA).^2 + (dCD_dA*dA).^2 + (dCD_dq*dq).^2 );

if strcmpi(tag,'none'), label = 'none (baseline)';
else, label = tag;
end
curve = struct('family',fam,'tag',tag,'label',label,'Re',Re, ...
        'alpha',alpha(:),'CL',CL(:),'CD',CD(:), ...
        'dCL',dCL(:),'dCD',dCD(:), ...
        'q',q,'rho',rho,'S',S,'PkPa',PkPa,'TC',TC);

curves = [curves; curve]; %#ok<AGROW>

% Per-curve CSV
outcsv = fullfile(odir_out, sprintf('G7_%s_Re%d.csv', tag, round(Re)));
```

```
        writetable(table(curve.alpha, curve.CL, curve.dCL, curve.CD, curve.dCD, ...
          'VariableNames',{'alpha_deg','CL','dCL','CD','dCD'}), outcsv);
      end

      if ~isempty(curves)
        SETS.(fam)(r).Re = Re;
        SETS.(fam)(r).curves = curves;
      end
    end
end

%% Build appendix tables A-1, A-2, A-3

% Table A-1: Density and Uncertainty

if exist('Tcond','var') && ~isempty(Tcond)
    namesC2 = san_vars(Tcond.Properties.VariableNames);
    iCfg2 = find(contains(namesC2,{'wingletconfig','config','winglet'}),1,'first');
    iT2 = find(contains(namesC2,{'temp','temperature'}),1,'first');
    iP2 = find(contains(namesC2,{'pressure','pkpa'}),1,'first');

    cfg_str = string(Tcond{:,iCfg2}); % Winglet Config
    p_atm_kPa = Tcond{:,iP2}; % Pressure [kPa]
    T_C = Tcond{:,iT2}; % Temp [°C]

    nrow = numel(p_atm_kPa);
    dp_kPa = 0.1*ones(nrow,1); % machine error [kPa]
    dT_C = 1.0*ones(nrow,1);  % machine error [°C]

    Rgas = 287; % [J/kg·K]
    p_Pa = p_atm_kPa*1000; % [Pa]
    T_K = T_C + 273.15; % [K]
    dp_Pa = dp_kPa*1000; % [Pa]
    dT_K = dT_C; % [K]

    rho  = p_Pa ./ (Rgas.*T_K); % [kg/m^3]
    drho = sqrt( ((1./(Rgas.*T_K)).*dp_Pa).^2 + ...
            ((-p_Pa./(Rgas.*T_K.^2)).*dT_K).^2 );

    TableA1 = table(cfg_str, p_atm_kPa, T_C, dp_kPa, dT_C, rho, drho, ...
      'VariableNames',{'Config','p_atm_kPa','T_C','dp_kPa','dT_C','rho_kgm3','drho_kgm3'});
end

writetable(TableA1, fullfile(odir_out,'Table_A1_density_uncertainty.csv'));

% Table A-2: CL vs alpha plot data
A2_fam = {};
A2_tag = {};
```

```
A2_Re = [];
A2_alpha = [];
A2_CL = [];
A2_dCL = [];

fams = fieldnames(SETS);
for ii = 1:numel(fams)
    fam = fams{ii};
    for r = 1:numel(SETS.(fam))
        Re = SETS.(fam)(r).Re;
        curves = SETS.(fam)(r).curves;
        for k = 1:numel(curves)
            npts = numel(curves(k).alpha);
            A2_fam = [A2_fam; repmat({fam},npts,1)]; %#ok<AGROW>
            A2_tag = [A2_tag; repmat({curves(k).tag},npts,1)]; %#ok<AGROW>
            A2_Re = [A2_Re; Re*ones(npts,1)]; %#ok<AGROW>
            A2_alpha = [A2_alpha; curves(k).alpha(:)]; %#ok<AGROW>
            A2_CL = [A2_CL; curves(k).CL(:)]; %#ok<AGROW>
            A2_dCL = [A2_dCL; curves(k).dCL(:)]; %#ok<AGROW>
        end
    end
end
TableA2 = table(A2_fam, A2_tag, A2_Re, A2_alpha, A2_CL, A2_dCL, ...
    'VariableNames',{'Family','Config','Re','alpha_deg','CL','dCL'});
writetable(TableA2, fullfile(odir_out,'Table_A2_CL_vs_alpha.csv'));

% Table A-3: CL vs CD plot data
A3_fam = A2_fam;
A3_tag = A2_tag;
A3_Re = A2_Re;
A3_CL = A2_CL;
A3_dCL = A2_dCL;

A3_CD = [];
A3_dCD = [];

for ii = 1:numel(fams)
    fam = fams{ii};
    for r = 1:numel(SETS.(fam))
        curves = SETS.(fam)(r).curves;
        for k = 1:numel(curves)
            A3_CD = [A3_CD;  curves(k).CD(:)];  %#ok<AGROW>
            A3_dCD = [A3_dCD; curves(k).dCD(:)]; %#ok<AGROW>
        end
    end
end

TableA3 = table(A3_fam, A3_tag, A3_Re, A3_CL, A3_dCL, A3_CD, A3_dCD, ...
```

```
    'VariableNames',{'Family','Config','Re','CL','dCL','CD','dCD'});
writetable(TableA3, fullfile(odir_out,'Table_A3_CL_vs_CD.csv'));

assignin('base','TableA1',TableA1);
assignin('base','TableA2',TableA2);
assignin('base','TableA3',TableA3);

disp('Table A-1: Density and Uncertainty');
disp(TableA1);
disp(' ');
disp('Table A-2: CL vs Alpha Plot Data');
disp(TableA2);
disp(' ');
disp('Table A-3: CL vs CD Plot Data');
disp(TableA3);

%% Plot
make_plots_with_errors(SETS, odir_fig);

end

function make_plots_with_errors(SETS, odir)
palette = [1 0 0; 0 0 1; 0 0 0; 0 0.5 0];  % [red; blue; black; green]

fams = fieldnames(SETS);
for ii = 1:numel(fams)
    fam = fams{ii};
    for r = 1:numel(SETS.(fam))
        Re = SETS.(fam)(r).Re;
        curves = SETS.(fam)(r).curves;
        if isempty(curves), continue; end

        % CL vs alpha
        figure('Color','w');
        hold on;
        grid on;
        box on;
        for k = 1:numel(curves)
            col = palette(mod(k-1,size(palette,1))+1,:);
            x = curves(k).alpha;
            y = curves(k).CL;
            dy = curves(k).dCL;
            h = errorbar(x, y, dy, 'o-','LineWidth',1.5, 'DisplayName',curves(k).label);
            set(h,'Color',col,'MarkerFaceColor',col);
        end
        xlabel('\alpha (deg)');
        ylabel('C_L');
        title(sprintf('C_L vs \\alpha — %s family — Re = %d', fam, round(Re)));
```

```
    legend('Location','northwest');
    saveas(gcf, fullfile(odir, sprintf('CL_vs_alpha__%s__Re%d.png', fam, round(Re))));

    % Drag polar
    figure('Color','w');
    hold on;
    grid on;
    box on;
    for k = 1:numel(curves)
      col = palette(mod(k-1,size(palette,1))+1,:);
      x = curves(k).CD;
      dx = curves(k).dCD;
      y = curves(k).CL;
      dy = curves(k).dCL;

      % vertical error bars
      h = errorbar(x, y, dy, 'o-','LineWidth',1.5, 'DisplayName',curves(k).label);
      set(h,'Color',col,'MarkerFaceColor',col);

      % horizontal error bars
      yl = ylim;
      cap = 0.01*(yl(2)-yl(1));
      for i = 1:numel(x)
        if ~isfinite(x(i)) || ~isfinite(y(i)) || ~isfinite(dx(i)), continue;
        end
        line([x(i)-dx(i), x(i)+dx(i)], [y(i), y(i)], ...
            'Color',col,'LineWidth',1.5,'HandleVisibility','off');
        line([x(i)-dx(i), x(i)-dx(i)], [y(i)-cap, y(i)+cap], ...
            'Color',col,'LineWidth',1.5,'HandleVisibility','off');
        line([x(i)+dx(i), x(i)+dx(i)], [y(i)-cap, y(i)+cap], ...
            'Color',col,'LineWidth',1.5,'HandleVisibility','off');
      end
    end
    xlabel('C_D');
    ylabel('C_L');
    title(sprintf('Drag Polar (C_L vs C_D) — %s family — Re = %d', fam, round(Re)));
    legend('Location','northwest');
    saveas(gcf, fullfile(odir, sprintf('CL_vs_CD__%s__Re%d.png', fam, round(Re))));
  end
end
end
```

VII. **Acknowledgements**

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VIII. **References**

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